return type in Java

1. What is Return Type?

Return type in Java defines the type of value a method sends back after execution.

Example

```
int add(int a, int b) {
   return a + b;
}
```

Here:

- int → return type
- Method returns an integer value

🔹 Another Example

```
String getName() {
   return "Omii";
}
```

✔ Return type = String
✔ Returned value = "Omii"

🔹 Key Points

✔ Defines method output type
✔ Must match the returned value
✔ Helps reuse method results
✔ void means no return

2. Types of Return Types

1. Primitive Return Type

What is it?

When a method returns basic data types

Examples:

Int , long , char , boolean

Example

```
int getNumber() {
    return 100;
}

double getPrice() {
    return 99.99;
}
```

Returns single value
Stored directly

2. Non-Primitive Return Type

- **What is it?**

When a method returns an object or reference

Examples:

Example :

String , Arrays , Class Objects

```
String getName() {
    return "Omii";
}
```

3. void Return Type

"void in Java indicates that a method performs an action but does not return any value."

Method returns nothing

```
void showMessage() {
    System.out.println("Hello Java");
}
```

Return Statement

Used to send value back to the caller

```
return value;
```

Example:

```
int square(int num) {
    return num * num;
}
```

When to Use void?

- ✔ When you don't need result
- ✔ When method performs action only
- ✔ When output is directly shown (print)

4. Important Rules

Return type must match value
Wrong:

```
int getData() {
    return "Hello";
}
```

Correct:

```
String getData() {
    return "Hello";
}
```

void method cannot return value
Wrong:

```
void test() {
    return 10;
}
```

✔ Correct:

```
void test() {
    return;
}
```

Multiple return statements allowed

```
int check(int num) {
    if (num > 0) {
        return 1;
    } else {
        return -1;
    }
}
```

5. Memory Level Understanding

"When a method returns an object in Java, it returns the reference stored in stack, pointing to an object created in heap memory."

When a method is called:

1. It goes into stack memory
2. Executes the code
3. return sends value back
4. Method removed from stack

```
int result = sum(2, 3); // result = 5
```

Key Points

- ✔ Objects are always created in heap memory
- ✔ References are stored in stack memory
- ✔ Method returns reference, not actual object
- ✔ Local variables (like s) are destroyed after method ends
- ✔ Object survives if any reference exists
- ✔ Garbage Collector removes unused objects

6.Advanced Concept (Returning Object)

“Returning an object in Java means a method returns a reference of a class instance, allowing multiple data values to be accessed together."

```
class Student {
    String name;
}

Student getStudent() {
    Student s = new Student();
    s.name = "Omii";
    return s;
}
```

Key Points

- A method can return an object of a class
- Return type must be the class name
- Java returns reference, not actual object copy
- Object is created in heap memory
- Useful for returning multiple values together
- Helps in real-world modeling (Student, Employee, etc.)
- Common in frameworks (Spring, APIs, etc.)